Do Now · retrieval

Lesson 131 · Asymmetric Encryption and Public-Key Use

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What is the principal challenge of symmetric encryption between distant users?

Q2

What is a function parameter?

Q3

What is the binary result of 1 + 1 in a full adder with carry-in 0?

Q4

How many bits are in three bytes?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Generate key pair — Public key can be shared; private key remains secret.

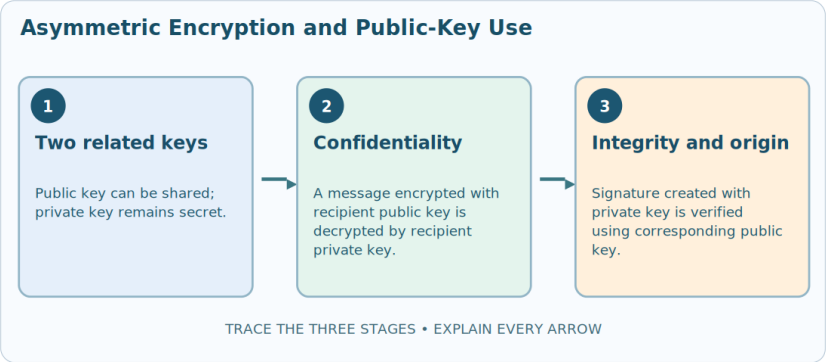
Core idea 2

Encrypt to recipient — A message encrypted with recipient public key is decrypted by recipient private key.

Core idea 3

Sign and verify — Signature created with private key is verified using corresponding public key.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

public key — A key that can be distributed openly.

private key — A matching secret key kept by its owner.

asymmetric encryption — Uses a related public/private key pair.

digital signature — Cryptographic evidence of origin and integrity with private-key signing.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

Alice wants to send Bob a private message using public-key encryption. Identify both keys used.

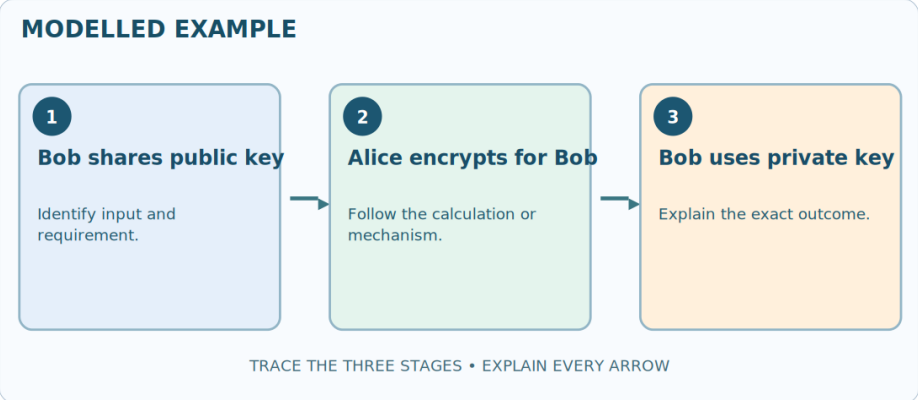
Follow the worked method

1. Bob shares public key — Predict the first step from the prompt.

2. Alice encrypts for Bob — Explain the intermediate result or decision.

3. Bob uses private key — Alice encrypts a suitably sized message or session key using Bob's public key. Bob decrypts using his matching private key. Alice never requires Bob's private key.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Publisher signs privately
Predict the first step from the prompt.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Explain the difference between public and private keys and how they support confidentiality. [6]

Question 2

Outline how a digitally signed download is verified. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Compare asymmetric and symmetric approaches for sending a large encrypted video file. [8]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

A public key must be kept secret for public-key encryption to work.

Use key vocabulary from memory

1. Explain public key in your own words.

2. Explain private key in your own words.

3. Explain asymmetric encryption in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.